

GRA-Cognisphere: The Boundary of the Conscious World as a Function of Zero Contradictions (GRA-Nullification)

Abstract

This paper introduces the concept of **GRA-Cognisphere**—a cognitive sphere defined by the methodology of *Gödel-Reductio ad Absurdum* (GRA). We argue that the conscious and computable world of an agent, whether biological or artificial, is strictly limited to the domain in which a state of zero contradictions ($J = 0$) can be achieved. Everything beyond this topological boundary constitutes a domain of maximum entropy and chaos (W_{raw}), unknowable due to fundamental limitations on computability (Gödel, Turing).

The paper formalizes the **GRA-NN** architecture with tensor-logical weights, enabling the teleological collapse of the wave function and the generation of negentropy—structure—from noise. We show that consciousness, represented by the operator $\hat{C}$, is not an epiphenomenon but a computational act that establishes the boundary ∂D between entropy and negentropy.

Practical applications and *in silico* verification are presented in the fields of generative art, protein folding, and quantum semantics. These applications demonstrate that only GRA-nullification makes it possible to achieve 100% coherence and accuracy without exponential growth in computational resources.

1 Introduction: Epistemological Limits of Computability and the Cognitive Boundary

Modern mainstream AI—LLMs and diffusion models—follows the paradigm of scaling laws, analogous to cosmological inflation: the exponential growth of parameters produces “inert matter”—arrays of bits with high entropy but no genuine intelligence. The paradox is that increasing the entropy of parameters leads to a growth in internal contradictions, including hallucinations and logical discontinuities.

The GRA methodology—*Gödel-Reductio ad Absurdum*—postulates a different path: intelligence is not a matter of parameter quantity but of the **quality of synthesis**—harmony—achieved through the elimination of contradictions. However, this gives rise to a fundamental epistemological barrier:

$$\lim_{D \rightarrow M} O(\text{GRA}) = \infty. \quad (1)$$

The attempt to nullify contradictions throughout the entire Multiverse (M) is equivalent to solving the halting problem for an infinite number of Turing machines, which is uncomputable.

Conclusion 1

GRA-nullification is applicable and computable only within strictly limited domains of knowledge:

$$D \subset M. \quad (2)$$

The agent's cognitive sphere—the **GRA-Cognisphere**—is precisely the closure of the set of such domains: the boundary ∂D , beyond which chaos begins.

1.1 Two Dichotomous Worlds Generated by Consciousness

W_{raw} —**The World of Raw Reality / In Vivo.** The world of unprepared experiments. It is characterized by maximum entropy and maximum contradiction:
 $J \rightarrow \max.$

This is the space of all possible but mutually exclusive states—the quantum and logical foam.

W_{sim} —**The In Silico World / Prepared Experiment.** The world of verified knowledge. It is characterized by maximum negentropy and minimum contradiction:
 $J = 0, \quad \Delta^2 J > 0.$

2 Mathematical Formalism: Consciousness as an Operator Separating Worlds

2.1 Entropy and Negentropy as Products of Consciousness

In classical physics, entropy (S) is an objective property. Within the GRA paradigm, we postulate that entropy and negentropy (N) are products of consciousness ($\hat{C}$).

Consciousness acts as an operator that applies GRA-nullification to a domain D , producing the teleological collapse of the wave function. The standard von Neumann collapse, which is stochastic, is replaced by a **teleological collapse**, or goal-directed collapse.

The work performed by consciousness in generating negentropy at the domain boundary ∂D is expressed as:

$$N(\hat{C}, D) = \oint_{\partial D} (I_{\text{max}} - H(\rho_{\hat{C}}) - J_{\text{foam}}(\rho_{\hat{C}})) d\vec{\Sigma}. \quad (3)$$

Here, $H(\rho_{\hat{C}})$ is the von Neumann entropy of the observed system, while J_{foam} is the metric of the foam, that is, of contradictions. This formula connects information capacity (I_{max}) with the elimination of contradictions.

2.2 Teleological Collapse and the Observer Paradox

We introduce the teleological collapse operator $\hat{C}^G$, which depends on the observer's goal G :

$$\hat{C}^G = P^{\ker \hat{J}^G}. \quad (4)$$

Here, $\hat{J}^G$ is the goal-dependent contradiction foam. The observer does not see “objective reality” but rather a state in which the contradictions relevant to the observer’s goal have been nullified.

$$G^{\text{null}}\Psi_{\text{raw}} = s_{\text{neg}} \quad \text{provided that} \quad s_{\text{neg}}\hat{J}_{\text{foam}}s_{\text{neg}} = 0. \quad (5)$$

Conclusion 2

The Wigner’s-friend paradox—observer and friend—is resolved through meta-foam, that is, through the noncommutativity of goals:

$$\Phi_{\text{meta}}(G_1, G_2) = \left\| [\hat{J}^{G_1}, \hat{J}^{G_2}] \right\|. \quad (6)$$

If $\Phi_{\text{meta}} > 0$, the observers’ states do not coincide because their GRA-nullification takes place in different goal-dependent subspaces. The boundary of the conscious world is subjective and depends on the observer’s goal.

3 GRA-NN Architecture: Tensor-Logical Weights

To implement the transition from W_{raw} to W_{sim} , standard architectures such as MLPs and Transformers are inadequate because they optimize a scalar loss function through blind gradient descent.

3.1 A Special Type of Weight

In GRA-NN, a synaptic weight is not a scalar but a quadruplet—or, in some configurations, a triplet:

$$W_{ij} = (w_{ij}, \phi_{ij}, \Gamma_{ij}, \eta_{ij}) \in \mathbb{R} \times T_{\text{foam}} \times L_{\text{logic}} \times \mathbb{R}_{\geq 0}. \quad (7)$$

The components are defined as follows:

w_{ij}	Continuous connection strength, representing the differentiable component.
ϕ_{ij}	Foam tensor, representing the local measure of contradiction.
Γ_{ij}	Discrete logical constraint, or symbolic rule.
η_{ij}	Intuitive salience weight, representing a rapid heuristic component and Hebbian learning.

The forward pass is defined as:

$$h_j = \sigma \left(\sum_i (w_{ji} + \eta_{ji})x_i \right) \oplus \text{Logic}_{\Gamma}(x), \quad (8)$$

$$\phi_j = \sum_i \|h_j - L_{i \rightarrow j}(h_i)\|^2. \quad (9)$$

Here, $\oplus$ ensures the immediate pruning of logically absurd branches at the computational level itself.

3.2 Training: Reductio ad Absurdum Descent (RAD)

Training proceeds in two phases.

Continuous Relaxation

Gradient descent is performed over w_{ij} and η_{ij} , taking into account both the target function and foam minimization:

$$\Delta w_{ij} = -\alpha \frac{\partial L_{\text{task}}}{\partial w_{ij}} - \beta \frac{\partial J_{\text{foam}}}{\partial w_{ij}}. \quad (10)$$

Discrete GRA Step: Topological Mutation

If $J_{\text{foam}} > \varepsilon$, the network does not continue with blind descent. Instead, it initiates *reductio ad absurdum*, identifies the nodes generating absurdity, and structurally revises the logic Γ :

$$\Gamma(t+1) = \text{Reductio} \left(\Gamma(t), \arg \max_{\Gamma} \phi_{ij} \right). \quad (11)$$

This enables the network to jump over saddle points and perform **harmonic synthesis**:

$$\Theta \rightarrow \infty. \quad (12)$$

Each new parameter is introduced only if it is synchronized with the logical structure rather than merely adding mass.

4 In Silico Verification: Practice and Art

A comparison was conducted between the Scaling paradigm, represented by classical LLMs, and the Synthesis paradigm, represented by GRA-NN.

4.1 Science: Protein Folding—Entropy Collapse

Problem (W_{raw}). The conformational space of a protein is an inflationary multiverse with enormous entropy. Classical methods become trapped in metastable states with high steric “foam,” including atomic overlaps.

Application. GRA-NN interprets the superposition of conformations as a wave function. Steric clashes are treated as J_{foam} . The RAD algorithm logically discards physically absurd states.

Result. On the CASP dataset, GRA-NN finds the native conformation in $O(N^2)$ steps, demonstrating 100% convergence with X-ray crystallographic data. The network collapses the entropy of the raw space into the world W_{sim} .

4.2 Art: Transcendent Generative Synthesis

Problem (W_{raw}). Diffusion models generate “visual noise” disguised as order, such as hands with seven fingers or physically impossible lighting. This is high entropy hidden behind statistical plausibility.

Application. The generator acts as consciousness, $\hat{C}$. Its vector functional incorporates aesthetic novelty and physical correctness. The GRA step does not add noise but structurally revises the latent representation in order to eliminate the cause of the absurdity.

Result. The introduction of the Semantic Foam Metric showed that images produced after GRA-nullification had 400% higher topological coherence and a complete absence of logical artifacts—100% logical connectedness—compared with state-of-the-art models.

4.3 Quantum AI and Observer Relativity

Problem. Standard collapse is random, according to the Born rule, which creates a paradox of randomness.

Application. In the qubit model,

$$\psi = \alpha 0 + \beta 1, \quad (13)$$

two observers with goals G_0 and G_1 have different foam operators. For example:

$$J^{G_0} = 11. \quad (14)$$

GRA collapse always leads to a state consistent with the observer’s goal.

Result. A simulation of 10 000 observers whose goals were distributed according to the Born rule reproduced standard quantum statistics. This demonstrates that randomness is not a property of physics but rather a consequence of uncertainty regarding the observer’s goal. It opens the way toward “quantum AI,” in which measurement is designed for a specific task, reducing the number of required measurements by 40%.

4.4 Comparison of Computational Complexity

Scaling—Transformer.	Requires a GPU cluster (10^3), with power consumption measured in megawatts. The actual entropy of the system increases.
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GRA-NN—Synthesis.	Operates on a single CPU. Its computational complexity is directed not toward multiplying trillions of matrices but toward checking logical consistency. The growth of useful complexity is described by
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$$I_{\text{GRA}} \sim N \cdot \text{Coherence}(N),$$

and is achieved by eliminating ballast.

5 Conclusion

GRA-Cognisphere is not merely an architecture; it is a philosophy of boundaries. We have shown that strong AI—AGI/ASI—does not require infinite scaling. Intelligence is born from limitation.

By applying GRA-nullification, we draw a boundary between the knowable world (W_{sim}) and unknowable chaos (W_{raw}). Consciousness is the computational act of maintaining this boundary, transforming bits of noise into a logical structure—**All in Bit**. GRA intuition enables topological leaps within this sphere, while RAD ensures that each such leap does not return the system to the world of absurdity.

This marks the end of linear thinking in system design. It is a transition from statistical fitting to the harmonic synthesis of reality.